

Note that each survivor will get \$131.25 as her total return on the original investment of \$100. The 31.25% investment return contains 5% of the bank's money and a healthy 26.25% of something we call "mortality credits." These credits represent the capital and interest "lost" by the deceased and "gained" by the survivors.

The catch, of course, is that the average nonsurvivor forfeited her claim to the funds. And while the beneficiaries of the nonsurvivor might be frustrated with the outcome, the survivors get a superior investment return. Personally, I find no other financial product that guarantees such high rates of return, conditional on survival.

More importantly, they *all* get to manage their lifetime income risk in advance, without having to worry about what the future will bring. This is the essence of the benefit from immediate annuities.

In fact, this story can be taken one step further. What if the group decided to invest the \$500 in the stock market, or some risky NASDAQ high-tech fund, for the next year? Moreover, what happens if this fund or subaccount collapses in value during the next year and falls 20% in value to \$400? How much will the surviving members lose? Well, if you are thinking "nothing," that is absolutely the correct answer. They divide the \$400 among the surviving four and get their original \$100 back.

Such is the power of mortality credits. They subsidize losses on the downside and enhance gains on the upside. In fact, I would go so far as to say that once you wrap true longevity insurance around a diversified portfolio, the annuitant can actually afford and tolerate more financial risk.

Of course, real live annuity contracts do not work in the way described previously. The group's "tontine" contract is renewable each year and the surviving 96-year-olds have the option to take their mortality credits and go home. In practice, annuity contracts are for life and these credits are spread and amortized over many years of retirement. The annuitant receives a constant periodic check that blends all of these varying components. But the basic insurance economics underlying the contract are exactly as described earlier.

While this life-roulette game would not yield such high returns at younger ages—and one might be better off managing the money oneself with a systematic withdrawal plan—by the mid-'80s beating the